

# Alberta Opioid Mortality Around the COVID-19 Onset

An interrupted time series analysis using federal data

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Independent project

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[github.com/Adilbek123/alberta-substance-use-trends](https://github.com/Adilbek123/alberta-substance-use-trends)

# Question to be Answered

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- Did Alberta's opioid death rate shift sharply when COVID-19 hit, beyond what the prior trend explains?
  - Cutoff used: 2020 Q2, the first full quarter under Alberta's public health emergency
  - Outcome: opioid-related deaths per 100,000 Alberta residents per quarter
- Why this matters for Mental Health and Addiction work
  - The post-COVID baseline shapes what we measure as 'success' under the Alberta Recovery Model
  - Understanding the size of the shift is a prerequisite for any decomposition or follow-up policy work

# Background and Context

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- Alberta opioid mortality from 2016 to early 2020
  - Quarterly opioid death rate sat around 4 per 100,000
  - Flat trend over the five years prior to COVID — no sustained rise or fall
- March 2020: Alberta declares a public health emergency
  - Many things changed at once — virus circulation, public health measures, harm reduction services, the unregulated drug supply, mental health services, social and economic conditions
- Federal data shows opioid mortality rose sharply across Canada from 2020 onward
  - This analysis quantifies the Alberta-specific change at the cutoff

# Approach

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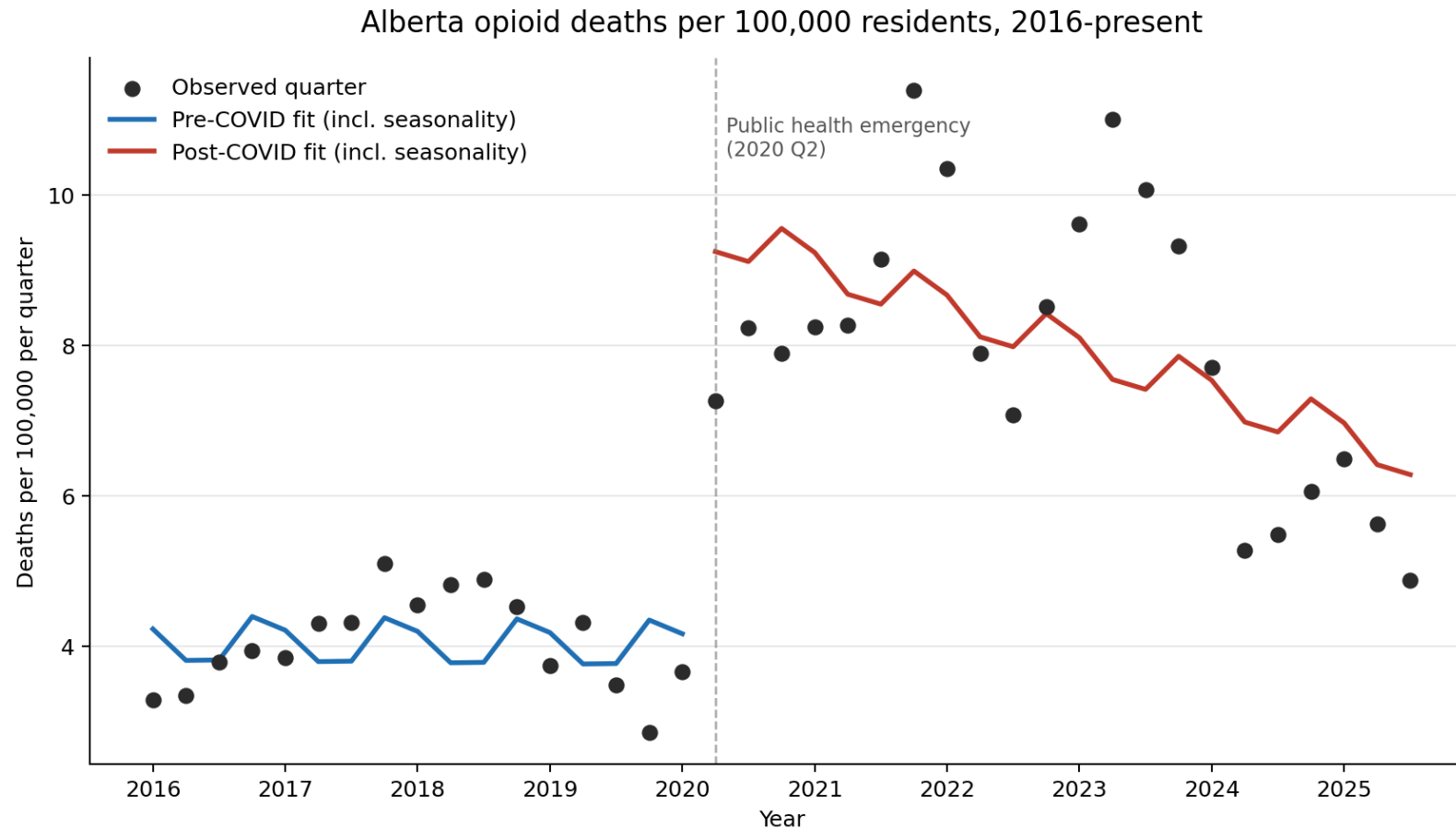
- **Interrupted time series**
  - Compare the trend in the years before the cutoff to the trend in the years after
  - Measure the gap between them at the cutoff itself
- **Adjustments to keep the comparison fair**
  - Use a rate per 100,000 residents (not raw counts), so population growth does not drive the result
  - Adjust for seasonality — opioid mortality has a predictable quarter-by-quarter pattern
- **Robustness checks to ensure the result is not fragile**
  - Move the cutoff back into the pre-period as a placebo — should produce no jump
  - Drop the quarter that straddles the cutoff and refit
  - Re-fit using a count model with population offset as a cross-check

# Data

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- Public Health Agency of Canada — Substance-Related Harms Data
  - Quarterly opioid deaths in Alberta
  - 2016 Q1 through 2025 Q3 — 39 quarters in total
  - Downloaded 20 May 2026 from [health-infobase.canada.ca](https://health-infobase.canada.ca)
- Statistics Canada Table 17-10-0009-01 — Quarterly population estimates
  - Used to convert death counts into a rate per 100,000 residents
- No internal Government of Alberta data used
  - All sources are public and reproducible

# Findings — A Sharp, Large, Persistent Increase



- Pre-COVID average
  - About 4 deaths per 100,000 per quarter
- Post-COVID average
  - About 8 deaths per 100,000 per quarter
- Level shift at 2020 Q2
  - About 5.5 per 100,000 per quarter
  - Roughly a doubling of the rate
  - Statistically very strong
- Pre-COVID trend was flat
  - The jump is not a continuation of an existing trend

Sources: PHAC Health Infobase substance-related harms data; StatCan Table 17-10-0009-01.

Level shift at 2020 Q2 (segmented regression with quarter FE, HAC SE 3-lag, small-sample correction): +5.50 per 100k [95% CI +3.60, +7.39].

# Robustness — The Result Is Not Fragile

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- Placebo cutoffs in the pre-period (2018–2019)
  - When we move the cutoff to a date before COVID, the model finds no comparable jump
  - Largest placebo movement is about a third of the real jump, and in the opposite direction
- Dropping the quarter that straddles the cutoff (2020 Q1)
  - The estimate moves by less than three per cent
- Different statistical assumptions about standard errors
  - The estimate is the same; the size of the confidence interval barely changes
- A count-model cross-check
  - The post-COVID rate is about 2.4× the pre-COVID rate, consistent with the main estimate

# What This Tells Us

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- There is a real, sharp, sustained change in Alberta opioid mortality at the COVID onset
  - Roughly a doubling of the rate, holding pre-trend and seasonality constant
  - The change is large enough to persist for years, with the rate only returning toward baseline since 2024
- The change is attributable to the COVID-19 pandemic broadly
  - This includes the virus, public health measures, supply chain disruption, harm reduction service changes, mental health service disruption, social isolation, and economic shock
  - These are all downstream consequences of the pandemic — they came together by construction, not by coincidence

# What This Does Not Tell Us

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- We cannot separate the virus channel from everything else that came with the pandemic
  - What share came from supply toxicity vs. service capacity vs. isolation vs. economic shock is a separate question
  - Answering it needs different data — drug-checking composition, service utilization, etc.
- A peer-province comparison would not isolate a clean 'COVID-only' effect
  - All Canadian provinces faced COVID and its policy response at essentially the same time
  - Comparing Alberta to BC or Ontario would identify Alberta-specific deviation, not the virus channel alone
- The estimate captures the level shift at the cutoff, not the full trajectory
  - The post-COVID period peaked in 2021 Q4 and has been declining since 2024 — the average effect varies year by year

# Next Steps

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- Mechanism decomposition
  - Bring in supply-side data, service-capacity data, and outcome data to estimate the share of the jump attributable to each channel
- Heterogeneity
  - Break the result down by age, sex, geographic zone, and substance type (fentanyl, carfentanil, other)
- Post-period dynamics
  - Model the rise-to-peak-and-decline shape directly, rather than collapsing it into a single level shift
- Cumulative excess deaths
  - Report a policy-relevant total alongside the immediate level shift

# Questions?

Code, data, and full documentation: [github.com/Adilbek123/alberta-substance-use-trends](https://github.com/Adilbek123/alberta-substance-use-trends)